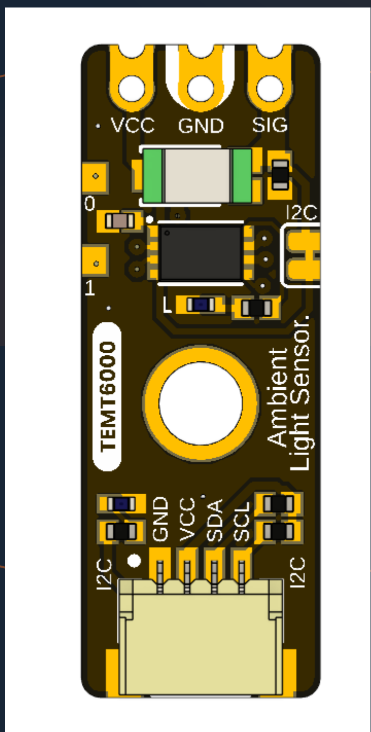
PRODUCT REFERENCE MANUAL

UNIT ELECTRONICS · DEVLAB ECOSYSTEM · ATOM FAMILY

UNIT ATOM TEMT6000

Version 1.1.0



TEMT6000 ambient-light module with an onboard I2C interface, Qwiic expansion, and direct analog access.

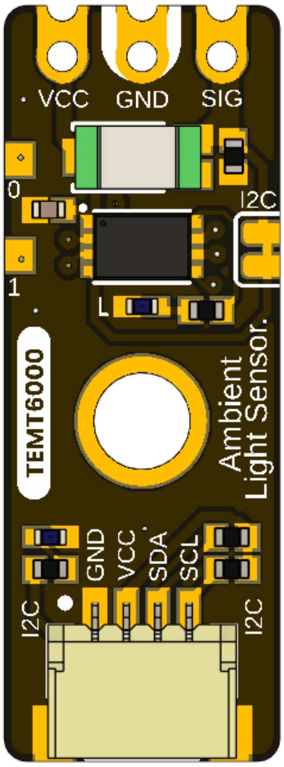
TEMT6000 | I2C + ANALOG | QWIIC | DEVLAB ECOSYSTEM | ATOM FAMILY

DOCUMENT MAP

Contents

Description

The UNIT ATOM TEMT6000 is an ambient-light sensor module developed by UNIT Electronics as part of the DevLab ecosystem. It combines a TEMT6000 visible-light phototransistor with an onboard interface controller, providing both I²C communication and direct access to the sensor's analog signal.



The board provides two alternative Qwiic connection positions, direct VCC, GND, and SIG contacts, reserved PA0 and PA1 pads, power and built-in status indicators, and a solder bridge for disabling the I²C interface. The controller reads the sensor through its PA2 ADC input. The built-in indicator is controlled through PB5, while the I²C interface uses PB6/SCL and PA10/SDA. These pins are also shared with the factory programming interface as SWCLK and SWDIO.

Applications

- I2C-connected ambient-light acquisition
- Direct analog light measurement during development and calibration
- Automatic display and indicator brightness control
- Day/night and relative-light detection
- Environmental data logging
- Educational I2C, ADC, and phototransistor experiments

DevLab Format Compatibility

As part of the DevLab Atom family, the TEMT6000 module uses a compact, standardized format for rapid prototyping and integration into sensing systems. It provides direct access to the analog signal through the VCC, GND, and SIG contacts, while the onboard controller publishes digital sensor readings through I²C. The module is intended for nominal 3.3 V or 5 V operation. The controller guidance uses a conservative 2.0–5.5 V range; complete-module electrical limits require the current schematic and validation.

The board has two optional positions for horizontal 4-pin, 1.00 mm-pitch JST/Qwiic connectors carrying GND, VCC, SDA, and SCL. They provide alternative I²C connection points. Before connection, confirm the populated connector and its orientation, and ensure the host tolerates the actual I²C pull-up voltage; a 3.3 V host may require level translation when the module is powered at 5 V. The pull-up resistance is not specified by the released documentation.

Hardware Features

- TEMT6000 ambient-light phototransistor
- 32-bit Arm Cortex-M0+ controller with 16 KB Flash and 2 KB SRAM, using its internal HSI at up to 24 MHz
- Nominal 3.3 V and 5 V operation; controller upper operating limit 5.5 V
- 7-bit I2C slave operation from 100 kHz through 400 kHz
- Qwiic GND, VCC, SDA, and SCL routing
- Direct analog SIG access with adjacent VCC and GND
- Reserved PA0 and PA1 pads with no current application assignment
- Internal sensor ADC connection to controller pin PA2
- Shared PB6 / SCL / SWCLK / PA10 / SDA / SWDIO controller mapping
- Factory-only SWD/reset functions, power LED, and PB5-controlled built-in LED
- Cuttable solder bridge for disabling I2C operation
- Manufacturer Part Number (MPN) UE0098

The controller implements DevLab Device Protocol (DDP) v1.0. The TEMT6000 profile is identified by Device ID `0x0102`; command `TEMT6000_RAW` (`0x80`) returns a 12-bit ADC sample in an unsigned 16-bit little-endian response. The current controller firmware reports factory address `0x20`, firmware/hardware 1.0, and capabilities `0x000001B9`.

`SWDIO` shares physical pin `PA10` with SDA, and `SWCLK` shares PB6 with `SCL`; there is no separate SWD port and the two modes are mutually exclusive. SWD/reset are reserved for manufacturer programming and advanced factory diagnostics, not user firmware replacement.

1 The Board

The design supports two host paths. A Qwiic-capable controller can attach through the I2C positions, while an ADC-capable host or test instrument can sample the exposed SIG contact directly. Factory debug reuses the same controller pins and physical port as I2C; it is not a separate user interface.

1.1 Accessories

No accessory bundle is specified. Typical integration items are:

Accessory	Purpose	Selection notes
Qwiic cable	Connects GND, VCC, SDA, and SCL	Verify 1.0 mm pitch, orientation, and contact order; connectors are shown as optional
I2C-capable host	Scans and communicates with the module	Use 7-bit addressing and a clock from 100 kHz through 400 kHz
Analog test lead or carrier	Accesses VCC, GND, and SIG	SIG must connect to a voltage-compatible ADC input
Logic analyzer	Checks I2C activity	Use input thresholds compatible with the powered board
Reference lux meter	Supports optical calibration	Required for quantitative illuminance validation

1.2 Board Identification

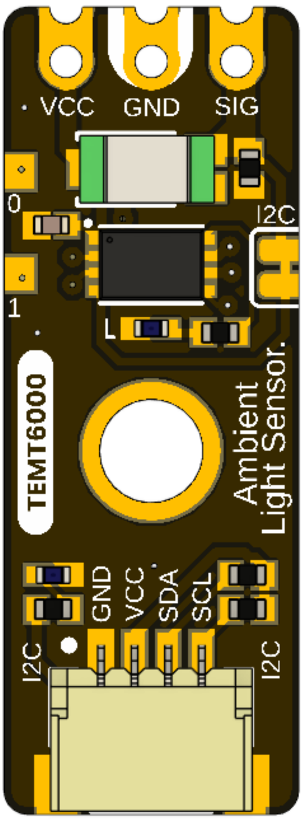
Item	Value
Product	UNIT ATOM TEMT6000 Ambient Light Sensor
Brand / company	UNIT Electronics
Board ecosystem	DevLab
Product family	Atom
Product type	I2C-compatible and direct-analog ambient-light module
Optical component	TEMT6000
Interface controller	32-bit Arm Cortex-M0+; 16 KB Flash and 2 KB SRAM; fitted suffix not asserted
Manufacturer Part Number (MPN)	UE0098
Current board artwork	
Available schematic	Legacy analog hardware V0.0.1 only
Product Reference	Version 1.1.0

Board, pinout, schematic, and documentation revisions are controlled independently.

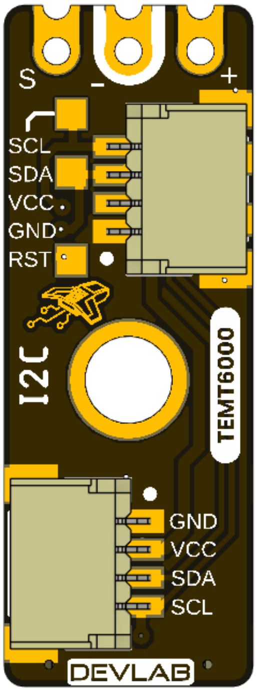
1.3 Identified Assemblies

Assembly	Function	Source status
TEMT6000 sensor	Converts visible light to photocurrent	Functional identity shown by board/pinout artwork; manufacturer and exact suffix unconfirmed
Interface controller	Samples or processes the sensor for I2C access	32-bit Arm Cortex-M0+; 16 KB Flash and 2 KB SRAM; exact fitted variant not confirmed; conservative controller voltage guidance is 2.0–5.5 V
Qwiic positions A and B	I2C power and bus access	Shown as optional horizontal JST connectors
Direct contacts	VCC, GND, and analog `SIG`	Identified on the top view
I2C disable bridge	Disconnects or disables I2C when cut	Function identified; exact circuit unspecified
Power and built-in LEDs	Power and firmware indication	`BUILTIN` is driven by controller PB5
PA0/PA1 and service functions	Reserved GPIO and factory reset/debug	`PA0`/`PA1` have no current application; SWD aliases share the I2C port
Internal controller mapping	PA2 ADC, PB5 built-in actuator, `PB6`/`SCL`/`SWCLK`, `PA10`/`SDA`/`SWDIO`	Current firmware/hardware mapping

1.4 Board Views



The top view shows the direct sensor contacts, controller, sensor, mounting hole, indicator circuitry, and one Qwiic connector position.



The bottom view labels Qwiic, the SWD aliases on that same port, reset, and the second optional connector position.

1.5 Handling

Use normal ESD precautions. Keep the transparent sensor package clean and optically unobstructed. Remove power before changing connectors or modifying the I2C solder bridge. Do not attach an SWD probe or attempt to replace the firmware. SWD/reset access is reserved for the manufacturer and advanced factory diagnostics; I2C must be inactive and isolated before factory SWD use.

2 Ratings

No current-revision schematic or released board-level electrical table is present for V0.3.1. This chapter therefore separates the expected Qwiic integration domain from comparative TEMT6000X01 reference data and explicitly marks current module values that still require release. That external data does not identify the manufacturer or exact part fitted to the board.

2.1 Current Module Operating Conditions

Parameter	Value	Status
Nominal module supply	3.3 V or 5 V	Supported operating points; use a current-limited source during bring-up
Controller operating-voltage guidance	2.0–5.5 V	Conservative controller-only range used until the exact fitted variant is confirmed; it includes nominal 3.3 V and 5 V operation
I2C logic domain	Follows the actual bus pull-up rail	At 5 V, use a 5 V-tolerant host or bidirectional level translation
I2C bus speed	100 kHz to 400 kHz	Supported operating range
I2C addressing	7-bit slave	Valid configurable addresses are `0x08..0x77`
Factory I2C address	`0x20` (7-bit)	Use `0x20` in I²C libraries. `0x40` is the 8-bit write address, not the device address.
Device protocol	DDP v1.0	Command transaction followed by an exact-length read transaction
Logical Device ID	`0x0102`	TEMT6000 identity; independent of I2C address
Firmware / hardware	1.0 / 1.0	Current observable controller profile
Capability bitmap	`0x000001B9`	I2C config, analog input, sensor data, relay, watchdog, persistent config
Raw digital sample	`0` to `4095`	12-bit ADC code returned as unsigned 16-bit little endian
ADC update interval	Approximately 20 ms	Background acquisition; I2C read returns latest published sample
Command processing delay	2 to 5 ms	Use 5 ms conservatively before reading
Pending setter timeout	250 ms	Parameter must arrive before expiry
Direct `SIG` range	Not specified	Requires current analog-stage schematic and measurement
Module current consumption	Not specified	Controller, LEDs, pull-ups, and sensor load are undocumented
Module ambient temperature	Not specified	No complete-module qualification supplied

The 2.0–5.5 V guidance applies only to the controller. It does not validate 5 V operation for every circuit on the module. The direct `SIG` path and I2C levels are referenced to the powered system, so every attached host must tolerate the resulting voltage or use level translation. Complete-module absolute maximums, pull-up resistance, and current consumption still require the schematic and qualification.

2.2 Interface Controller Characteristics

The supplied UE0102 reference manual identifies a PY32F003L24D6TR and publishes 1.7–5.5 V and −40 to +85 °C for that separate development board. It does not establish which ordering suffix was procured for this UE0098 assembly. This document therefore uses the narrower 2.0–5.5 V range as conservative guidance for the controller until the exact fitted variant is confirmed.

Feature	Controller capability
CPU and application clock	32-bit Arm Cortex-M0+; internal HSI at up to 24 MHz; external HSE oscillator not used
Memory	16 KB Flash and 2 KB SRAM
Controller operating-voltage guidance	2.0–5.5 V; conservative range used because the exact fitted variant is not confirmed
ADC	One 12-bit ADC, up to 10 external channels, conversion range `0..VCC`
I2C	Standard mode 100 kHz, Fast mode 400 kHz, 7-bit addressing
GPIO	Up to 18 I/Os, all available as external interrupts
Timers	TIM1, TIM3, TIM14, TIM16, TIM17, LPTIM, IWDG, WWDG, SysTick, IRTIM
Other interfaces	SPI, two USARTs, DMA, RTC, CRC-32, two comparators, UID; SWD is reserved for factory use on this module

The controller range applies to the controller only. It does not establish the pull-up resistance, complete-module current, or absolute maximum of every external contact.

2.3 TEMT6000 Maximum Ratings

The values below are included only as a comparative TEMT6000 profile. They are not proof that a Vishay part is fitted, are not guaranteed for the fitted sensor, and do not define limits for the controller, LEDs, pull-ups, connectors, or complete board.

Parameter	Symbol	Value	Unit
Collector-emitter voltage	`VCEO`	6	V
Emitter-collector voltage	`VECO`	1.5	V
Collector current	`IC`	20	mA
Power dissipation at 25 °C	`PV`	100	mW
Junction temperature	`Tj`	100	°C
Component operating temperature	`Tamb`	−40 to +100	°C

2.4 TEMT6000 Characteristics

Unless noted otherwise, the values are specified at 25 °C and are provided for comparison only. Production specifications must come from the confirmed fitted part and module-level validation.

Parameter	Test condition	Min.	Typ.	Max.	Unit
Collector dark current	`VCE = 5 V`, `Ev = 0`	—	3	50	nA
Collector light current	`Ev = 20 lx`, CIE illuminant A, `VCE = 5 V`	3.5	10	16	μA
Collector light current	`Ev = 100 lx`, CIE illuminant A, `VCE = 5 V`	—	50	—	μA
Collector-emitter capacitance	`VCE = 0 V`, `f = 1 MHz`, dark	—	16	—	pF
Collector-emitter saturation voltage	`Ev = 20 lx`, `IPCE = 1.2 μA`	—	0.1	—	V
Angle of half sensitivity	—	—	±60	—	degrees
Peak sensitivity wavelength	—	—	570	—	nm
Spectral bandwidth at half sensitivity	—	440	—	800	nm

The component values in Sections 2.3 and 2.4 are published by Vishay in document 81579. The complete source is listed in Chapter 8, Reference Documentation.

2.5 Legacy Analog Circuit Scope

The V0.0.1 schematic shows a TEMT6000 with a 10 kΩ emitter resistor, giving the first-order relation `VSIGNAL ≈ IPCE × 10 kΩ` outside saturation. Applying the 100 lx typical current from the datasheet predicts about 0.50 V; this is not a guaranteed current-board value.

V0.3.1 adds an interface controller and other circuitry. Without its schematic, the legacy resistor value and transfer function must not be represented as a guaranteed current-board characteristic. Direct `SIG` must be measured and validated on V0.3.1.

2.6 Unspecified Current-Module Characteristics

- Complete-module absolute maximum, current consumption, and power-up behavior
- Exact oscillator/timing tolerance and electrical bus-loading limits
- Pull-up resistance, bus capacitance allowance, and level compatibility
- Direct analog transfer function, load resistance, range, accuracy, and source impedance
- Guaranteed lux range, accuracy, repeatability, response time, and calibration
- Exact controller package/ordering suffix, released firmware image, and update procedure

- Board-level temperature, ESD, EMC, humidity, and ingress ratings

2.7 Electrical Precautions

1. Use a current-limited 3.3 V or 5 V supply for engineering bring-up.
2. Verify connector orientation and share ground before applying power.
3. Do not exceed the controller upper operating limit of 5.5 V.
4. At 5 V, confirm host tolerance at `SDA`, `SCL`, and `SIG`, or add suitable level translation before connection.
5. During normal operation, use the shared `PA10/SDA/SWDIO` and `PB6/SCL/SWCLK` lines only for I2C. SWD is an alternate factory function on these same physical lines, not a separate user interface.
6. Remove power before cutting or reworking the I2C solder bridge.
7. Confirm the current schematic and interface specification before production.

3 Functional Overview

The UNIT ATOM TEMT6000 combines an ambient-light phototransistor with an onboard controller. The module provides two ways to observe light level: digital readings over I2C and direct access to the analog sensor signal at `SIG`.

The TEMT6000 itself is an analog device. The onboard controller samples its output, applies the selected averaging, and provides the latest raw reading through DevLab Device Protocol (DDP) v1.0.

3.1 Functional Block Diagram

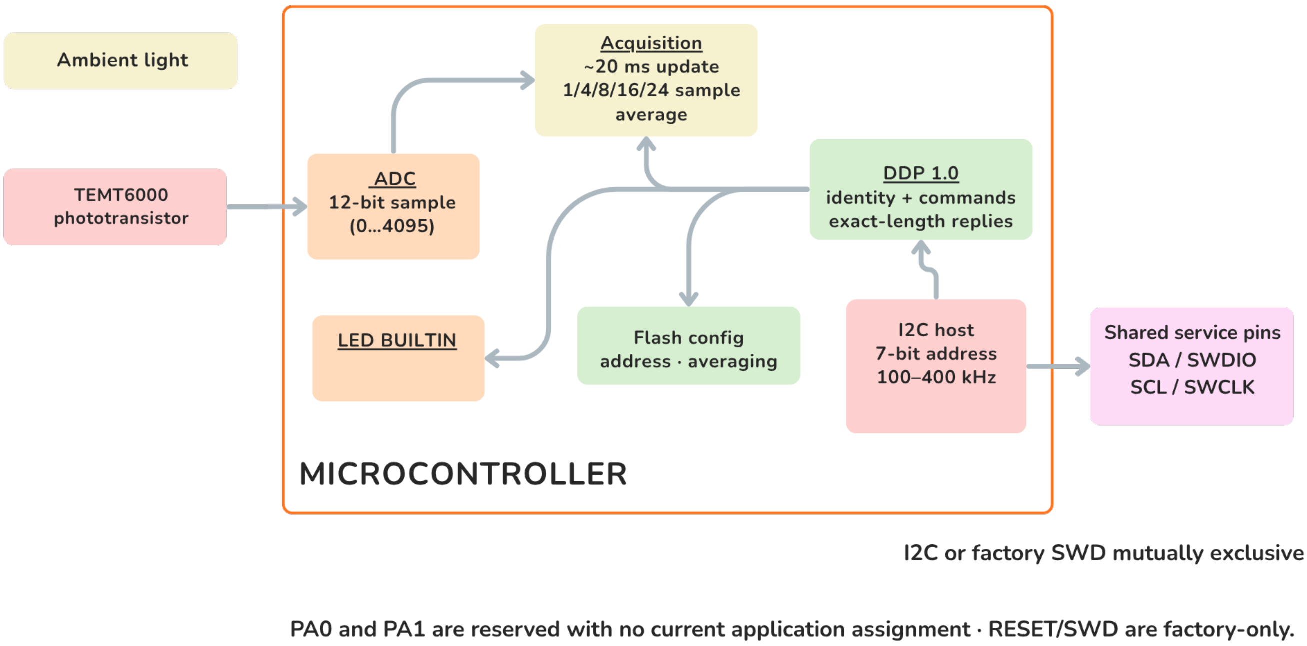


Figure 3.1 — Functional signal and control paths. Ambient light changes the TEMT6000 output. The signal is available at `SIG` and is also sampled through `PA2/ADC0` for access over I2C.

3.2 Operating Modes

Mode	Connection	User-visible result
Digital I2C	VCC, GND, SDA, and SCL	Averaged 12-bit raw light reading through DDP
Direct analog	VCC, GND, and `SIG`	Light-dependent voltage for a host ADC or measurement instrument

Both modes observe the same sensor path. An I2C read returns the latest available sample; it does not trigger a new light measurement. The direct analog output may be used independently for development, comparison, or application-specific calibration.

3.3 Digital Light Measurement

The onboard ADC produces an unsigned 12-bit value. A higher value represents a higher sensor output under the actual supply, optical, and mechanical conditions.

Measurement property	Value
ADC resolution	12 bits
Raw output range	`0..4095`
Response format	Two bytes, `[LSB, MSB]`
Update interval	Approximately 20 ms
Selectable averaging	1, 4, 8, 16, or 24 samples

The averaging setting controls the balance between response speed and stability:

Samples	Approximate time for a completely new average	Typical use
1	20 ms	Fast changes
4	80 ms	Low-latency smoothing
8	160 ms	General monitoring
16	320 ms	Stable ambient-light monitoring
24	480 ms	Maximum available smoothing

The result is a relative ADC reading, not a calibrated lux measurement. Quantitative lux measurements require calibration in the final enclosure and optical arrangement.

3.4 I2C Communication Model

The I2C host always initiates communication. The 7-bit I2C address selects the module, while the one-byte DDP selector identifies the requested function. These are independent values.

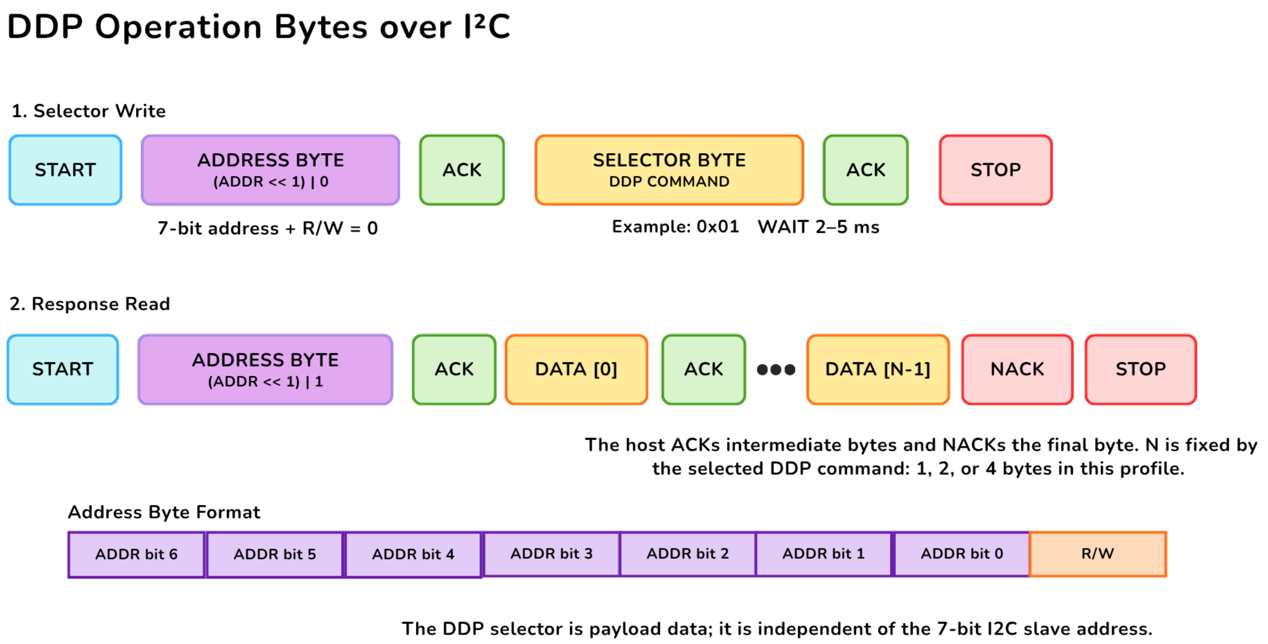


Figure 3.2 — DDP command and response over I2C. The host writes one DDP selector, issues STOP, waits 2–5 ms, and then reads the documented number of response bytes. The host acknowledges intermediate bytes and sends NACK after the final byte.

Communication property	Value
Factory 7-bit address	`0x20`
Configurable address range	`0x08..0x77`
I2C clock	100 kHz to 400 kHz
Delay before response read	2–5 ms; use 5 ms conservatively
Byte order for multi-byte values	Little endian
Supported response lengths	1, 2, or 4 bytes, depending on the selected function

Pass `0x20` directly to an I2C library. Values `0x40` and `0x41` are the shifted write/read address bytes visible on the bus, not the 7-bit address used by typical host APIs.

3.5 User-Accessible DDP Functions

User function	DDP selector	Result
Identify the module	`0x00`	Device ID `0x0102`

User function	DDP selector	Result
Read protocol version	0x04	DDP version 1.0
Read active I2C address	0x20	Current 7-bit address
Change I2C address	0x21	Stores and applies a new address in the valid range
Restore factory configuration	0x23	Restores default configuration; reset required
Read ambient-light data	0x60 or 0x80	Latest two-byte 12-bit raw sample
Set ADC averaging	0x62	Selects 1, 4, 8, 16, or 24 samples
Read ADC averaging	0x63	Returns the active averaging setting
Turn BUILTIN off/on	0xA0 / 0xA1	Drives the built-in indicator LOW/HIGH
Pulse BUILTIN	0xA2	Starts a non-blocking timed pulse
Set/read pulse time	0xA3 / 0xA4	Uses 1–40 units of 25 ms

The I2C address, averaging selection, and pulse duration are retained after a power cycle. Changing a stored value should be an occasional configuration operation rather than part of the normal measurement loop.

Detailed transaction handling is already implemented by the maintained C++ and MicroPython examples listed in Chapter 8.

3.6 Built-In Indicator

PB5/BUILTIN is a user indicator controlled through the DDP actuator functions. The shared DDP command names contain RELAY, but this product uses them for the built-in indicator and does not provide an electromechanical relay output.

The pulse duration is configurable from 25 ms to 1000 ms in 25 ms steps. A timed pulse runs without interrupting light measurements or I2C communication.

3.7 Physical Interface Notes

The SIG contact provides direct access to the analog sensor path. Its complete board-level transfer function, source impedance, and guaranteed output range are not specified for the current revision. Use a high-impedance input and verify the actual range before connecting it to a lower-voltage ADC.

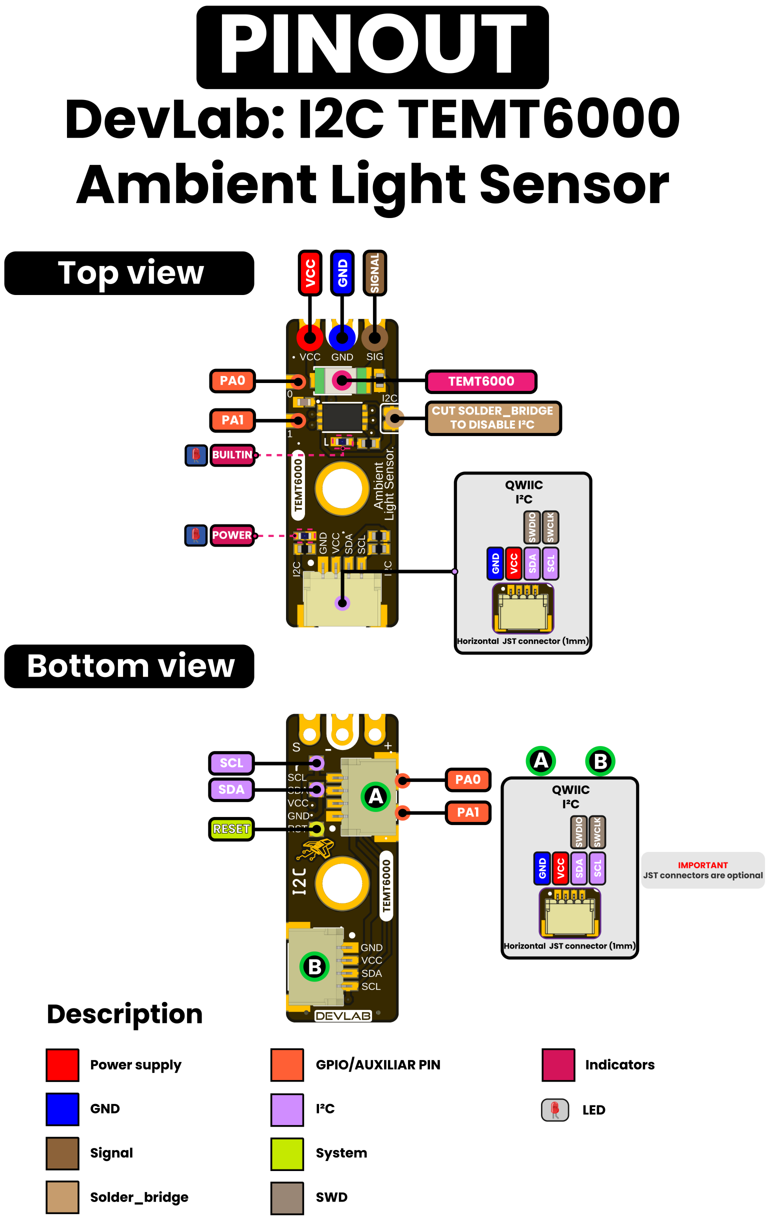
The normal I2C signals share controller pins with the service labels shown on the board: PA10 is SDA/SWDIO, and PB6 is SCL/SWCLK. There is no separate SWD connector. For normal product use, connect these lines as SDA and SCL.

PA0 and PA1 are reserved and have no current user function. The I2C disable bridge is intended for applications that require the analog path without active I2C operation; remove power before cutting or restoring it.

4 Connectors & Pinouts

Pin labels below come from the released English pinout. It does not provide controlled contact numbers or a mating-connector part number, so the signal order must be checked against connector orientation before producing a harness.

4.1 General Pinout



4.2 Direct Sensor Contacts

Label	Type	Description
VCC	Power	Module supply; nominal 3.3 V or 5 V operation
GND	Power	Common power and signal reference

Label	Type	Description
SIG / SIGNAL	Analog	Direct light-dependent sensor signal; current transfer function unspecified

These three contacts are shown at the end opposite the primary Qwiic connector.

4.3 Qwiic I2C Positions A and B

Signal	Type	Description
GND	Power	Common return
VCC	Power	I2C peripheral supply; nominal 3.3 V or 5 V
SDA / SWDIO	Bidirectional I2C / factory debug data	One physical PA10 line; normal use is SDA; host must tolerate the bus pull-up voltage
SCL / SWCLK	I2C clock / factory debug clock	One physical PB6 line; normal use is SCL at 100 kHz to 400 kHz

The pinout shows two possible horizontal 1.0 mm JST connector positions and marks JST connectors as optional. Verify which position is populated on the ordered assembly. Parallel connectors are intended for bus pass-through, but that topology must be confirmed by the current schematic.

I2C and SWD are multiplexed on this same physical port: PA10 is SDA/SWDIO, and PB6 is SCL/SWCLK. They are mutually exclusive, and there is no independent SWD connector or pin pair.

4.4 Auxiliary and Factory Service Functions

Label	Documented role	Qualification status
PA0	GPIO / reserved pin	No current application assignment; electrical limits unspecified
PA1	GPIO / reserved pin	No current application assignment; electrical limits unspecified
RESET	System reset	Factory service only; not intended for routine user operation

Do not connect an SWD probe or drive reset/debug functions. They are not user interfaces, and the product is not designed for user firmware replacement. Manufacturer diagnostics must make I2C inactive and isolate other bus devices before selecting SWD on the shared pins.

4.5 Indicators and I2C Bridge

The pinout identifies a POWER LED, a firmware-controlled BUILTIN LED, and a cuttable bridge used to disable I2C. Controller pin PB5 drives BUILTIN. Indicator polarity/current and bridge net connections still require the current schematic. The sensor ADC input is mapped internally to controller PA2; it is not an additional external contact in this pinout.

5 Board Operation

This chapter shows how to verify one UNIT ATOM TEMT6000 and read its raw ambient-light value with Arduino. The example supports the Pulsar ESP32-C6 and Pulsar RP2350 host boards.

5.1 Required Hardware and Software

- UNIT ATOM TEMT6000 module
- Pulsar ESP32-C6 or Pulsar RP2350 development board
- I2C connection for VCC, GND, SDA, and SCL
- USB cable for programming and Serial Monitor access
- Arduino IDE with the selected board package installed
- `DevLab_TEMT6000` Arduino library

5.2 Library Installation

1. Open **Tools > Manage Libraries** in Arduino IDE.
2. Search for `DevLab_TEMT6000`.
3. Select the current release and click **Install**.
4. Allow Arduino IDE to install the `DevLabDDP` dependency when prompted.

Figure 5.1 — DevLab_TEMT6000 in Arduino Library Manager. The installed package provides the TEMT6000 examples and installs the DDP communication dependency used by the sketch below.

5.3 I2C Connection and Example Settings

Connect the module with power removed.

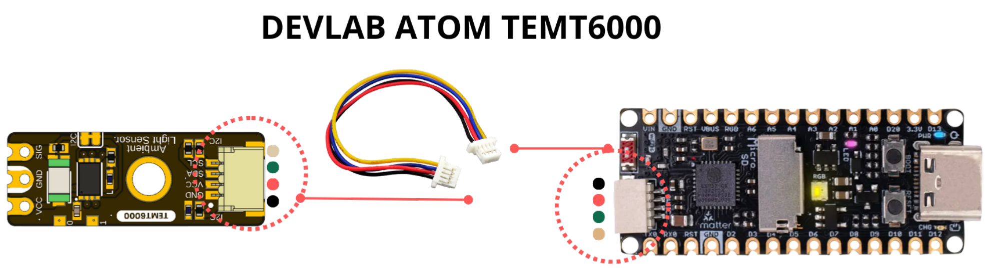


Figure 5.2 — Qwiic connection to a Pulsar ESP32-C6 host. Align the connector with the orientation shown and verify VCC, GND, SDA, and SCL before applying power.

Module	Pulsar ESP32-C6	Pulsar RP2350	Function
VCC	3.3 V or 5 V	3.3 V or 5 V	Module supply
GND	GND	GND	Common reference
SDA	GPIO6	GPIO24	I2C data
SCL	GPIO7	GPIO25	I2C clock

The pin numbers above match `singleSensor.ino`. Change `I2C_SDA` and `I2C_SCL` when the selected host board uses different I2C pins. At 5 V, confirm that the host accepts the bus pull-up voltage or use bidirectional level translation.

The example uses the following settings:

Constant	Default	Purpose
`I2C_SDA`	GPIO6 on Pulsar ESP32-C6; GPIO24 on Pulsar RP2350	Host SDA pin
`I2C_SCL`	GPIO7 on Pulsar ESP32-C6; GPIO25 on Pulsar RP2350	Host SCL pin
`I2C_FREQ`	400000 Hz	I2C clock
`SENSOR_ADDRESS`	`0x20`	Factory 7-bit module address
`READ_INTERVAL_MS`	1000 ms	Time between printed readings

If the module address was changed previously, update `SENSOR_ADDRESS` before uploading the sketch.

5.4 Single-Sensor Arduino Example

Open or create `singleSensor.ino`, then use the following sketch:

```
/**
 * @file singleSensor.ino
 * @brief Verifies a single TEMT6000 DDP device on the I2C bus and prints
 *        its raw ADC0 readings over Serial at a fixed interval.
 * @author Cesar Bautista
 */

#include <Arduino.h>
#include <Wire.h>
#include <DevLabDDP.h>
#include <DevLabI2CBusRecovery.h>

#if defined(ARDUINO_ARCH_RP2040)
#define I2C_BUS Wire
constexpr uint8_t I2C_SDA = 24U, I2C_SCL = 25U;
#elif defined(ARDUINO_ARCH_ESP32)
#define I2C_BUS Wire
constexpr uint8_t I2C_SDA = 6U, I2C_SCL = 7U;
#else
#error "Use Pulsar ESP32-C6 or Pulsar RP2350"
#endif

constexpr uint32_t I2C_FREQ = 400000;
constexpr uint8_t SENSOR_ADDRESS = 0x20;
constexpr uint32_t READ_INTERVAL_MS = 1000U;

DevLabDDP::Master master(I2C_BUS, DevLabDDP::DEVICE_TEMT6000);
bool deviceVerified = false;

void setup() {
  Serial.begin(115200);
  delay(500);

  if (!devlabBeginI2cBusRecovered(
          I2C_BUS, I2C_SDA, I2C_SCL, I2C_FREQ, 100)) {
    Serial.println("ERROR: I2C bus is blocked");
    return;
  }

  DevLabDDP::DeviceInfo info;
  deviceVerified = master.matchesExpectedDevice(SENSOR_ADDRESS, &info);

  if (!deviceVerified) {
    Serial.println("ERROR: address is not a DDP TEMT6000 (ID 0x0102)");
    return;
  }

  DevLabDDP::printDeviceInfo(
      Serial,
      SENSOR_ADDRESS,
      info,
      DevLabDDP::DEVICE_TEMT6000);

  Serial.println("adc0_raw");
}

void loop() {
  uint16_t raw;

  if (deviceVerified && master.readAdc(SENSOR_ADDRESS, 0, raw)) {
    Serial.println(raw);
  } else {
    Serial.println("ERR");
  }

  delay(READ_INTERVAL_MS);
}
```

5.5 Upload and Serial Output

1. Select the connected Pulsar ESP32-C6 or Pulsar RP2350 board in Arduino IDE.
2. Select its serial port.
3. Compile and upload `singleSensor.ino`.
4. Open Serial Monitor at **115200 baud**.
5. Illuminate and cover the TEMT6000 to confirm that the printed value changes.

At startup, the sketch recovers and initializes the I2C bus, verifies that address `0x20` contains a DDP TEMT6000 with Device ID `0x0102`, and prints the reported device information. It then prints the heading `adc0_raw` and one reading per second.

```
<DDP device information>
adc0_raw
<value from 0 to 4095>
<value from 0 to 4095>
```

The readings are raw ADC values. They indicate relative light level and must not be interpreted directly as lux.

5.6 Troubleshooting

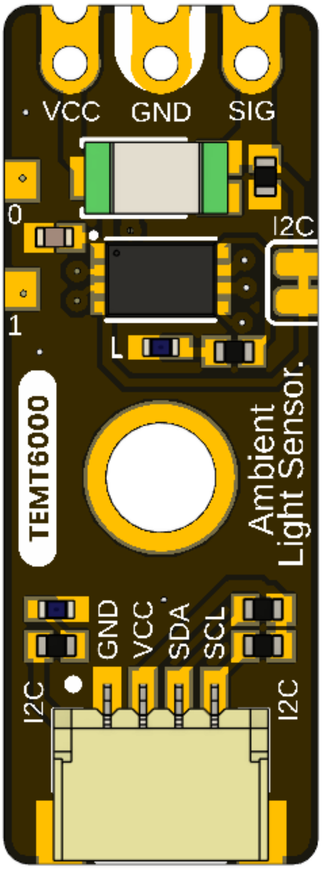
Serial output or symptom	Check
`ERROR: I2C bus is blocked`	Wiring, connector orientation, shared ground, pull-up voltage, and unintended SDA/SCL drive
`ERROR: address is not a DDP TEMT6000 (ID 0x0102)`	Address setting, I2C pins, module power, and possible address collision
ERR after successful identification	Cable continuity, bus clock, supply stability, and I2C connections
Constant `0` or `4095`	Sensor obstruction, excessive illumination, supply, and module orientation
Values change but lux is inaccurate	Calibrate the complete optical installation against a reference light meter

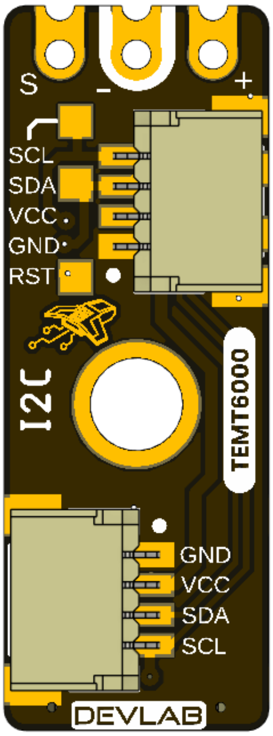
The complete Arduino examples and the reusable DDP library are listed in Chapter 8.

6 Mechanical Information

The current package provides rendered top and bottom views for V0.3.1 but no controlled current-revision dimension drawing. Legacy V0.0.1 dimensions are archived under `hardware/resources/old/` and describe a materially different, shorter analog-only board.

6.1 Current Board Envelope





The board has a narrow module outline, direct contacts at one end, at least one central mounting feature, and optional horizontal Qwiic connector positions. Rendered views are suitable for identification, not production measurement.

6.2 Integration Clearances

Leave optical clearance around the transparent TEMT6000 sensor package and its field of view. Provide connector insertion and cable bend clearance at every populated Qwiic position. Keep conductive fasteners away from pads and service signals, and avoid enclosure features that shadow the sensor or LEDs.

6.3 Mechanical Data Not Specified

- Board length, width, thickness, tolerances, and finished mass
- Hole diameters, coordinates, and edge clearances
- Optional connector population variants and mating-part numbers
- Connector insertion envelope, retention force, and cable bend radius
- Maximum top- and bottom-side component heights
- Optical axis, keep-out cone, and enclosure-window specification
- PA0/PA1 and service-pad coordinates

Do not scale production dimensions from the images or reuse the archived V0.0.1 mechanical values for V0.3.1.

7 Company Information

Item	Information
Company / brand	UNIT Electronics
Board ecosystem	DevLab
Product family	Atom
Website	UNIT Electronics
Address	Salvador 19, Cuauhtémoc, 06000 Mexico City, CDMX, Mexico

The product repository and controlled technical references are listed in the next chapter.

8 Reference Documentation

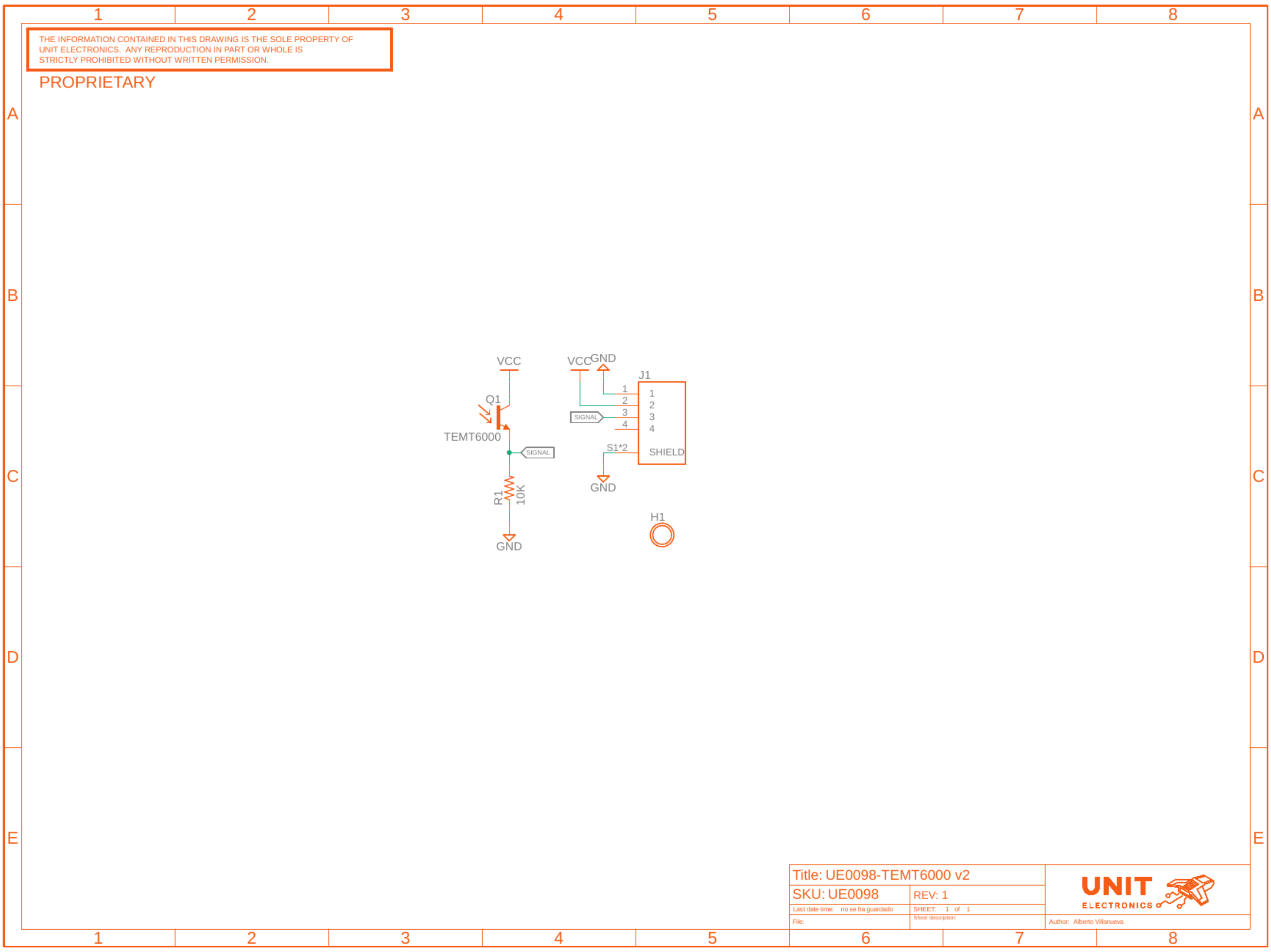
Reference	Location
Product repository	UNIT-Electronics-MX/unit_devlab_temt600_ambient_light_sensor
Hardware reference	hardware/README.md
English pinout	Released pinout — English
Spanish pinout	Released pinout — Spanish
Board top view	V0.3.1 resource
Board bottom view	V0.3.1 resource
Legacy schematic	Analog V0.0.1
Software examples	C++ and MicroPython
Arduino library	UNIT-Electronics-MX/unit_devlab_temt6000_library
Local DDP device profile	TEMT6000 DDP profile
Canonical DDP definitions and Arduino library	UNIT-Electronics-Labs/unit_devlab_ddp_library
Comparative component page	Vishay TEMT6000X01 — reference only
Source of component values in Sections 2.3 and 2.4	Vishay Semiconductors, *TEMT6000X01 Ambient Light Sensor*, document 81579 — reference only

The released pinout and current board images define the visible interfaces. The legacy schematic defines only the earlier analog module. The Vishay datasheet is retained only as a comparative TEMT6000X01 profile; it does not establish the manufacturer or orderable part fitted to this module and does not define complete-module ratings. The DDP profile defines runtime identity and digital data transport; it does not replace the missing V0.3.1 electrical specification.

9 Appendix

9.1 Legacy V0.0.1 Schematic

The image below is rendered from `hardware/unit_sch_V_0_0_1_ue0098_TEMT6000.pdf`. It documents the former analog-only module and is retained for traceability. It is **not** the electrical schematic for the controller-based V0.3.1 board.



The legacy circuit connects a TEMT6000 phototransistor and 10 kΩ load to one analog signal. It contains no interface controller, SDA, SCL, PA0, PA1, reset, SWD, indicators, or I2C disable bridge.

9.2 Document Control

Field	Value
Product	UNIT ATOM TEMT6000 Ambient Light Sensor
Product hierarchy	UNIT Electronics → DevLab board ecosystem → Atom family
Manufacturer Part Number (MPN)	UE0098
Current board artwork	V0.3.1

Field	Value
Current pinout	V3.1.0
Available schematic	Legacy V0.0.1 only
Product Reference	Version 1.1.0
Publication date	2026-08-31
Interfaces	Qwiic I2C with shared factory SWD, analog, and auxiliary pads

9.3 Required Technical Releases

- V0.3.1 schematic and bill of materials
- Exact controller package/ordering suffix, released firmware image, and update procedure
- Exact oscillator/timing tolerances and uniform future DDP status behavior
- Complete-module supply/logic absolute-maximum ratings and current consumption
- Pull-up values and bus-loading limits
- Direct `SIG` transfer, loading, range, and ADC guidance
- Current-revision controlled mechanical drawing
- Module-level optical, electrical, environmental, and EMC characterization

9.4 Source Notes and Inconsistencies

- Current board images and pinout resources use different revision-number formats in their filenames and storage directories.
- Current pinout artwork calls the product `DevLab: I2C TEMT6000`; this reference identifies UNIT Electronics as the company that creates and develops the DevLab board ecosystem, with Atom as a product family within it.
- The current pinout shows an onboard controller and I2C/debug resources, but the only schematic in the repository is the earlier analog V0.0.1 circuit.
- Archived V0.0.1 dimensions, topology, and board views do not describe the longer V0.3.1 controller-based board.
- Qwiic positions A and B are marked optional; released assembly variants and controlled contact numbering are not supplied.
- Earlier examples treated the sensor as a digital threshold input. The sensor signal is analog; I2C operation is provided by the new onboard controller.
- Current firmware maps PA2 to `ADC0` and PB5 to the relay-compatible actuator block. Digital-I/O commands `0x40..0x43` are not implemented; exposed `PA0`/`PA1` remain unassigned.

9.5 Current Firmware Limitations

- The `WATCHDOG` capability is announced, but the current watchdog command does not operate an active hardware IWDG and must remain experimental.
- `GET_RESET_INFO` is not implemented and NRST behavior is incomplete.
- An internal ADC error can be returned as `0x0FFF`, indistinguishable from a legitimate full-scale sample.
- Current main-loop code does not call the available prolonged-BUSY recovery.
- There is no sample timestamp, sample counter, or calibrated lux result.